

Test Plan: Car Mileage Validation for Fuel Expenses

1. Introduction

This test plan describes testing of mileage validation logic during fuel expense creation in the qaAuto application.

The goal is to verify that the system correctly allows or blocks expense creation depending on report date, car mileage, car creation date, existing expenses and future expenses.

2. Test items

The following functionality will be tested:

- Add an expense pop-up
- Report date field
- Mileage field
- Mileage validation rules
- Error messages for invalid mileage
- Car mileage update after successful expense creation
- Behavior with existing previous expenses
- Behavior with existing future expenses

3. Approach

Testing will be performed using functional and negative testing techniques.

The following scenarios will be covered:

- Creating the first expense
- Creating an expense for today
- Creating an expense for a past date
- Creating an expense when future expenses already exist
- Checking mileage against initial car mileage
- Checking mileage against previous expense mileage
- Checking mileage against next/future expense mileage
- Checking that invalid mileage values are blocked
- Checking that correct error messages are displayed
- Checking that car mileage is updated only when allowed

4. Entry / Exit criteria

Entry criteria

- User is logged in
- At least one car is added to Garage

- Add an expense pop-up is available
- Requirements/documentation are available
- Test environment is stable

Exit criteria

- All planned test cases are executed
- Passed/failed statuses are recorded
- Found bugs are documented
- Critical/blocking defects are reported
- Test results are reviewed

5. Environment needs

Testing environment:

- Web application: qaAuto
- Browser: Google Chrome
- OS: macOS / Windows
- Test data:
 - user account
 - car with initial mileage
 - expenses with different dates and mileage values
- Tools:
 - Jira for bug reports
 - TestCaseLab / Google Sheets for test cases
 - Confluence/design documentation for requirements

6. Responsibilities

Role	Responsibility
QA Engineer	Create and execute test cases
QA Engineer	Validate mileage logic and error messages
QA Engineer	Report found defects in Jira
Developer	Fix reported defects
Product Owner/Business Analyst/Teacher	Clarify unclear requirements

7. Risk and contingencies

Risk	Contingency
Mileage validation rules are unclear	Mark as requirement clarification / issue
Future expense logic is not fully described	Ask for clarification and document assumption
Test data may be hard to prepare	Create several cars/expenses with controlled dates
Error messages may differ from documentation	Report as documentation/UI inconsistency
Existing bugs may block testing	Report blocker and continue with unaffected scenarios